Osptek Display

TFT LCD SPECIFICATION

Model No:

YDP223B001-V9

RECORDS OF REVISION

Date	Rev.	Description	Note	Page
2022\3\31	**A**	**New sample**		

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1. SPECIFICATIONS

1.1 Features

Item	Standard Value
Display Type	200(R+G+B) *480Dots
LCD Type	a-Si TFT, Positive, Transmissive
Viewing Direction	IPS
Backlight	4 LED White Color
Interface	MIPI interface
Controller/driver IC	ST7797

1.2 Mechanical Specifications

Item	Standard Value	Unit
Outline Dimension	23.92(W) x57.79(L)x1.9(T)	mm
Active Area	21.72(W) x 52.42(L)	mm
Pixel pitch	/	mm

Note：For detailed information please refer to LCM drawing

1.3 Absolute Maximum Ratings

Item	Symbol	Condition	Min.	Max.	Unit
Power Supply Voltage	V_{DD}	-	-0.3	4.6	V
LCD Driver Supply Voltage	VGH-VSS	-	-0.3	18.5	V
Input voltage	Vin		-0.3	4.6	V
Operating Temperature	T_{OP}	-	-20	+70	°C
Storage Temperature.	T_{ST}	-	-30	+80	°C
Storage Humidity	H_D	Ta < 40 °C	-	90	%RH

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1.4 DC Electrical Characteristics

V_{DD} = 2.4~3.3V, V_{SS} = 0V, Ta = 25°C

Item	Symbol	Condition	Min.	Type	Max.	Unit
Logic Supply Voltage	V_{DD}	-	2.4	2.8	3.3	V
"H" Input Voltage	V_{IH}	-	0.8 V_{DD}	-	V_{DD}	V
"L" Input Voltage	V_{IL}	-	V_{SS}	-	0.2 V_{DD}	V
"H" Output Voltage	V_{OH}	-	0.8V_{DD}	-	V_{DD}	V
"L" Output Voltage	V_{OL}	-	V_{SS}	-	0.2 V_{DD}	V
Supply Current	I_{DD}	V_{DD} = 2.8V	-	4	6	mA

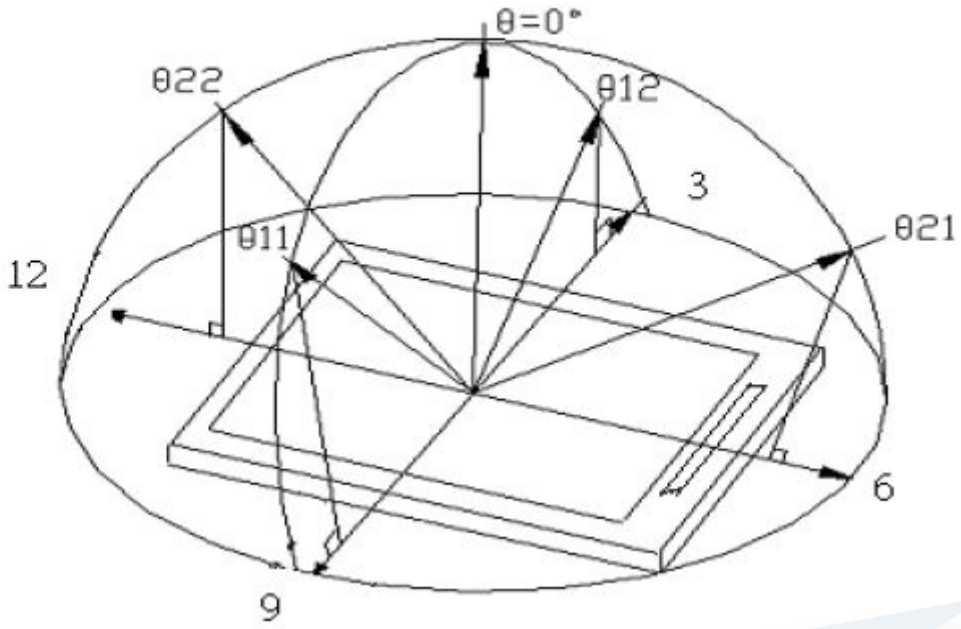
1.5 Optical Characteristics

Ta = 25°C

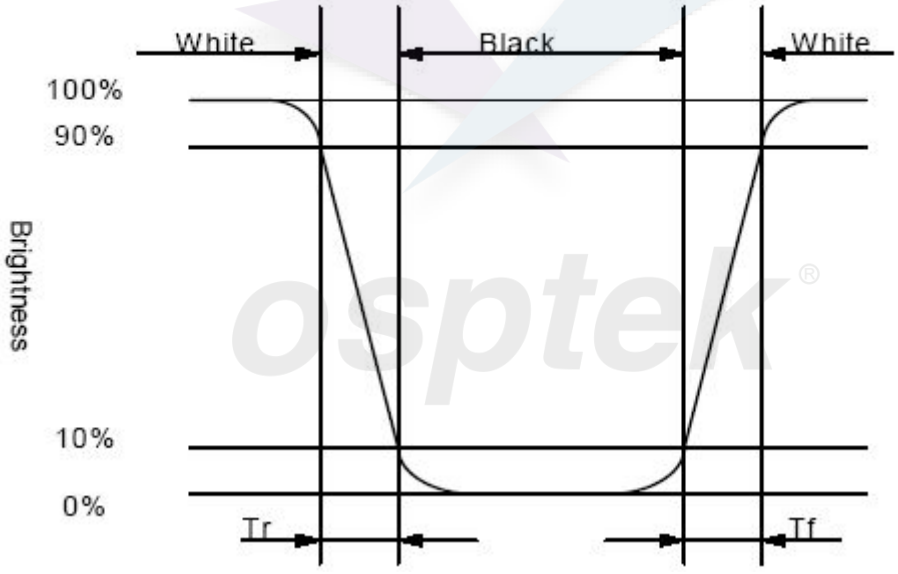
Item	Symbol	Conditions	Min.	Typ.	Max.	Reference
View Angle	θ11, θ12	C≥10,∅=0°	--	45	--	Note6-1
	θ21		--	45	--	Note6-1
	θ22		--	45	--	Note6-1
Contrast Ratio	C	θ=0°, ∅=0°	150	250	-	--
Response Time(rise)	tr	θ=0°, ∅=0°	-	10ms	20ms	Note6-3
Response Time(fall)	tf	θ=0°, ∅=0°	-	20ms	30ms	Note6-3
Luminance	B	θ =0° ∮ =0°	-		-	cd/m²

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Note 6-1 : The definitions of viewing angles



Note 6-3 : The definition of response time :



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1.6 Backlight & LED Characteristics

Maximum Ratings

Item	Symbol	Conditions	Min.	Max.	Unit
Forward Current	IF	Ta =25℃	-	20 (1 LED)	mA
Reverse Voltage	VR	Ta =25℃	-	5	V
Power Dissipation	PO	Ta =25℃	-	198	mW
Operating Temperature	T_{OP}	-	-20	70	℃
Storage Temperature	T_{ST}	-	-30	80	℃
Solder Temp. for 3 Seconds	-	-	-	260	℃

Electrical / Optical Characteristics **VSS = 0V, Ta =25℃**

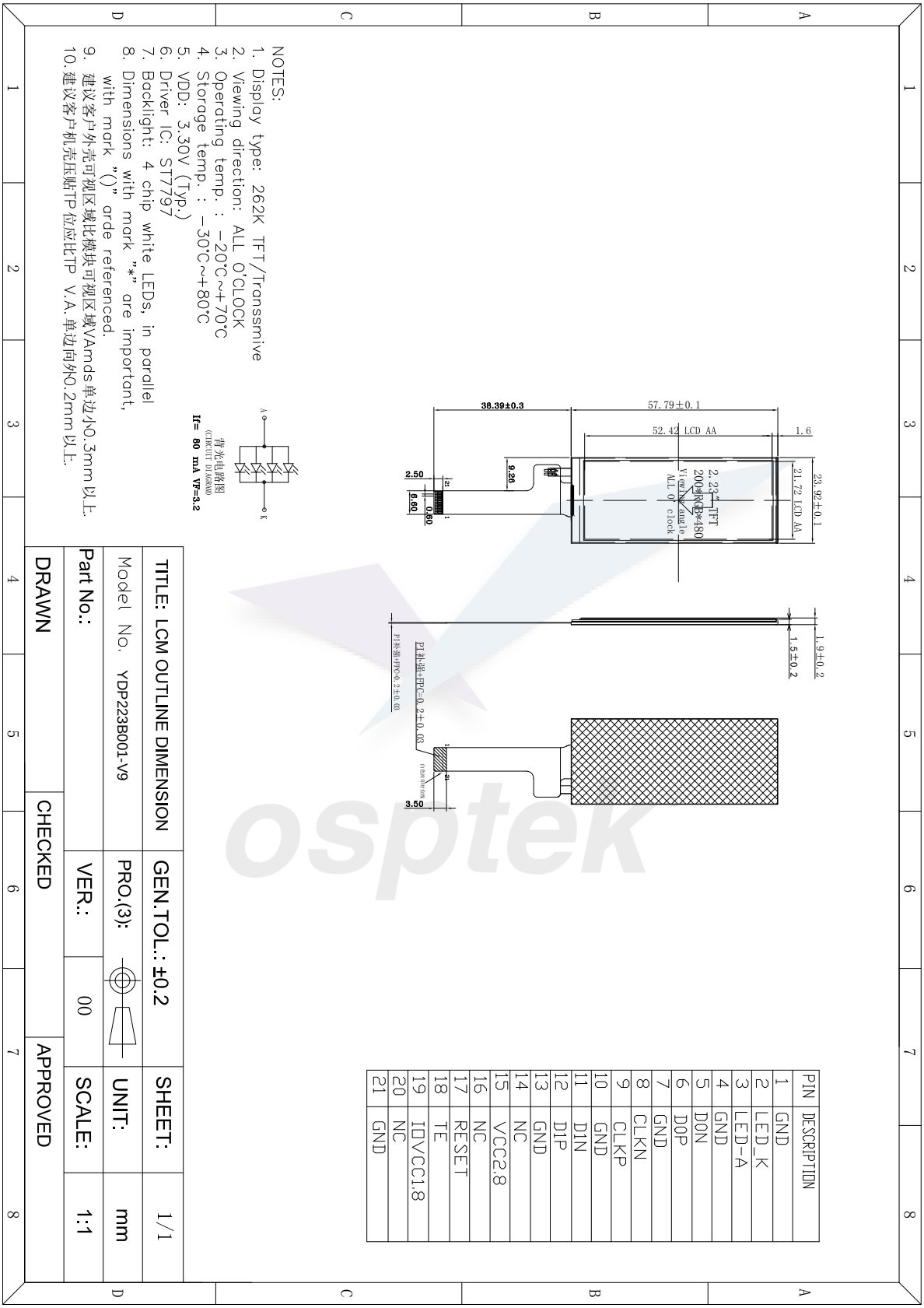
Item	Symbol	Conditions	Min.	Typ.	Max.	Unit
Forward Voltage	VF	IF= 80mA	2.8	3	3.2	V
Reverse Current	IR	-	-	-	-	uA
Average Brightness (without LCD)	IV	-	-	-	-	cd/m^2
CIE Color Coordinate (without LCD)	X	-	-	-	-	—
	Y		-	-	-	
Color	WHITE					

***1 This value will be changed while mass production.**

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2. MODU1LE STRUCTURE

2.1 Counter Drawing



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Interface Pin Description

Pin.No	Symbol	Function
1	GND	Power ground
2	LEDK	Power supply for LED. [Cathode]
3	LEDA	Power supply for LED. [Anode]
4	GND	Power ground
5	D0N	Differential data pairs for MIPI interface.
6	D0P	Differential data pairs for MIPI interface.
7	GND	Power ground
8	CLK_N	Differential clock or strobe pair for MIPI interfaces.
9	CLK_P	Differential clock or strobe pair for MIPI interfaces.
10	GND	Power ground
11	D1N	Differential data pairs for MIPI interface.
12	D1P	Differential data pairs for MIPI interface.
13	GND	Power ground
14	NC	No connect
15	VDD2.8V	Power supply （2.8V）
16	NC	No connect
17	RESET	Reset signal
18	TE	TE
19	VDD1.8V	Power supply （1.8V）
20	NC	No connect
21	GND	Power ground

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2.3 Timing Characteristics

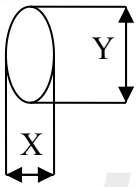
Please refer to ST7701S DATASHEET.

2.4 Display Command

Please refer to ST7701S DATASHEET.

3. INSPECTION SPECIFICATIONN

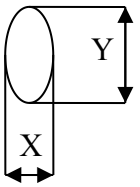
NO.	项目 Item	经验标准 Inspection Standard	判断 Result	备注 Note
1	整体功能 All functional defects	1）不显示 No display 2）显示异常 Display abnormally 3）缺划（横或竖，横&竖）Missing vertical, horizontal segment 4）短路 Short circuit 5）背光不亮或闪烁 Backlight no lighting, flickering and abnormal lighting.	不允许 Reject	
2	缺失 Missing	少成分 Missing component	不允许 Reject	
3	外观尺寸 Outline dimension	同 CD 图 Overall outline dimension beyond the drawing is not allowed		

NO.	项目 Item	检验标准 Inspection Standard	备注 Note
4	清楚的黑白点 Clear Spots	ϕ=（X+Y）/2  A：AA 区（显示区） B：VA 区（可视区） C：可视区以外(Out of VA)	

区域 Zone / 尺寸 Size	接受个数 Acceptable Quantity		
	A	B	C
ϕ≤ 0.1mm	Ignore		Ignore
0.1mm<ϕ≤0.2mm	3		
0.2mm<ϕ≤0.25mm	2		
ϕ>0.25mm	0		

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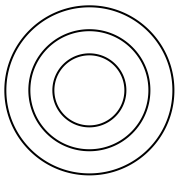
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<tr><th>NO.</th><th>项目
Item</th><th>检验标准
Inspection Standard</th><th>备注
Note</th></tr>
<tr><td>5</td><td>不明显的黑白点
Dim Spots</td><td>
ϕ=（X+Y）/2

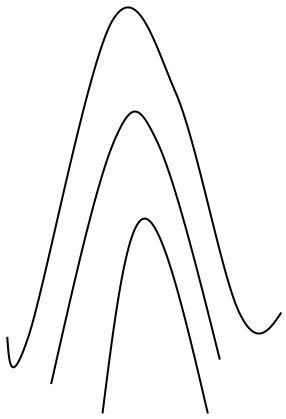
A：AA 区（显示区）
B：VA 区（可视区）
C：可视区以外(Out of V.A.)


<table>
<tr><th rowspan="2">区域 Zone
尺寸 Size</th><th colspan="3">接受个数
Acceptable Quantit</th></tr>
<tr><th>A</th><th>B</th><th>C</th></tr>
<tr><td>ϕ≤ 0.3mm</td><td colspan="2">Ignore</td><td rowspan="3">Ignore</td></tr>
<tr><td>0.3mm<ϕ≤0.6mm</td><td colspan="2">2</td></tr>
<tr><td>ϕ>0.6mm</td><td colspan="2">0</td></tr>
</table>
</td><td></td></tr>
<tr><td>6</td><td>线不良
Line defect</td><td>
<table>
<tr><th colspan="2">尺寸 Size（mm）</th><th colspan="3">接受个数 Acceptable Quantity</th></tr>
<tr><th>L（Length）</th><th>W（width）</th><th>A</th><th>B</th><th>C</th></tr>
<tr><td>Ignore</td><td>W≤0.03</td><td colspan="2">Ignore</td><td rowspan="3">Ignore</td></tr>
<tr><td>L〈5.0</td><td>0.03〈W≤0.05</td><td colspan="2">2</td></tr>
<tr><td></td><td>0.05〈W</td><td colspan="2">以脏污论
Define as spot defect</td></tr>
</table>
</td><td></td></tr>
<tr><td>7</td><td>偏光片刮伤
Polarizer Scratch</td><td>
<table>
<tr><th colspan="2">尺寸 Size（mm）</th><th colspan="3">Acceptable Quantity</th></tr>
<tr><th>L（Length）</th><th>W（width）</th><th>A</th><th>B</th><th>C</th></tr>
<tr><td>Ignore</td><td>W≤0.03</td><td colspan="2">Ignore</td><td rowspan="3">Ignore</td></tr>
<tr><td>L≤10</td><td>0.03〈W≤0.05</td><td colspan="2">2</td></tr>
<tr><td>L〈5.0</td><td>0.05〈W≤0.08</td><td colspan="2">1</td></tr>
<tr><td></td><td>0.08〈W</td><td colspan="2">0</td></tr>
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</td><td></td></tr>
</table>

<table>
<tr><td>8</td><td>偏光片与玻璃间气泡
Polarize Air bubble</td><td>
<table>
<tr><td rowspan="2">区域 Zone
尺寸 Size</td><td colspan="3">接受个数
Acceptable Quantity</td></tr>
<tr><td>A</td><td>B</td><td>C</td></tr>
<tr><td>φ≤ 0.2mm</td><td colspan="2">Ignore</td><td rowspan="4">Ignore</td></tr>
<tr><td>0.2mm<φ≤0.3mm</td><td colspan="2">2</td></tr>
<tr><td>0.3mm<φ≤0.5mm</td><td colspan="2">1</td></tr>
<tr><td>φ>0.5mm</td><td colspan="2">0</td></tr>
</table>
</td><td></td></tr>
</table>

牛顿环/干涉纹 Newton Ring

NO.	项目 Item	检验标准 Inspection Standard	备注 Note
9	规则 Inerratic	1. 在整个触摸屏检查区域内（可视区）超过 1/3 范围，不可；When Newton ring dimension is more than 1/3 of sample dimension, it is regarded as a defect. 2. 直径≤5mm，且在整个触摸屏检查区（可视区）域小于 1/3 范围，不影响透过率及失真；不计 When Newton ring dimension is less than 1/3 of sample dimension is not affect font effect and line distortion under a ceiling fluorescent light, it is acceptable.	

10	不规则 Atactic	1. 在照明环境下牛顿环有影响清晰度和透过率，失真；不可。As long as Newton ring affects font effect and line distortion under a ceiling fluorescent light, it is regarded as a defect. 在整个触摸屏检查区域（可视区）内，超过 1/2，不可。φ≤10mm；不计。When φ≤10mm ,it is acceptable	

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4. PRECAUTION RELATING PRODUCT HANDLING

4.1 SAFETY

4.1.1 If the LCD panel breaks , be careful not to get the liquid crystal to touch your skin.

4.1.2 If the liquid crystal touches your skin or clothes , please wash it off immediately by using soap and water.

4.2 HANDLING

4.2.1 Avoid any strong mechanical shock which can break the glass.

4.2.2 Avoid static electricity which can damage the CMOS LSI—When working with the module, be sure to ground your body and any electrical equipment you may be using.

4.2.3 Do not remove the panel or frame from the module.

4.2.4 The polarizing plate of the display is very fragile. So , please handle it very carefully, Do not touch , push or rub the exposed polarizing with anything harder than an HB pencil lead (glass , tweezers , etc.)

4.2.5 Do not wipe the polarizing plate with a dry cloth, as it may easily scratch the Surface of plate.

4.2.6 Do not touch the display area with bare hands , this will stain the display area.

4.2.7 Do not use ketonic solvent & aromatic solvent. Use with a soft cloth soaked with A cleaning naphtha solvent.

4.2.8 To control temperature and time of soldering is 280 ± 10°C and 3-5 sec.

4.2.9 To avoid liquid (include organic solvent) stained on LCM.

4.3 STORAGE

4.3.1 Store the panel or module in a dark place where the temperature is 25°C ± 5°C and the humidity is below 65% RH.

4.3.2 Do not place the module near organics solvents or corrosive gases.

4.3.3 Do not crush, shake , or jolt the module.

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